

海通国际电子研究

全球化与本地化

蒲得宇 Jeff Pu jeff.dy.pu@htisec.com

夏雅梦 Alicia Xia alicia.ym.xia@htisec.com

周扬 Yang Zhou y.zhou@htisec.com

June 2024

This research report is distributed by Haitong International, a global brand name for the equity research teams of Haitong International Research Limited (“HTIRL”), Haitong Securities India Private Limited (“HSIPL”), Haitong International (Japan) K.K. (“HTIJKK”), Haitong International Securities Company Limited (“HTISCL”), and any other members within the Haitong International Securities Group of Companies (“HTISG”), each authorized to engage in securities activities in its respective jurisdiction. Please refer to the appendix for the Analyst Certification, Important Disclosures and Important Disclaimer.

观点、结论

- 宏观因素稳定，半导体周期向上，偏好AI、台积电、半导体设备和苹果

半导体周期回升

- 半导体周期多项顺风，先进制程UTR上行，AI驱动价值量和英特尔外包启动
- 全球WFE于2025年或达USD1000亿以上，国内迭加本地化趋势
- 存储的报价增速放缓，DRAM库存升高，HBM供需不确定性

AI从云到端

- 英伟达供应链订单强劲，后续题材为GB200供应链
- 仍偏好AI网络800G交换机、光模块于1.6T趋势

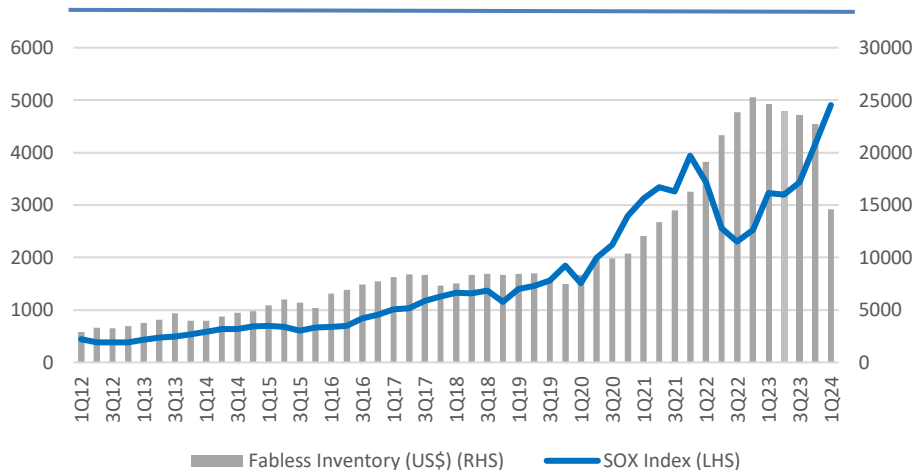
消费电子+AI

- 升评苹果，看好明后年产品周期 (+AI)
- SoC 藉AI提升价值量

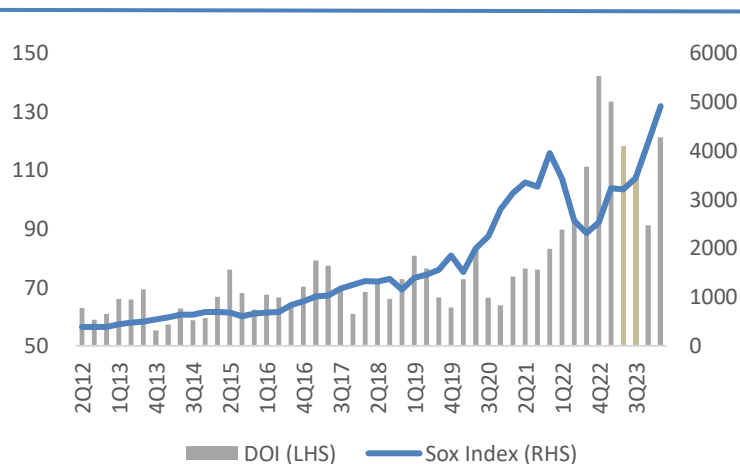
半导体库存维持健康水平

- 库存调整已大致完成，预期1季度库存保持较低水平

芯片设计公司库存金额 vs. 费半指数



芯片设计公司库存天数 vs. 费半指数



*1Q24尚未公告完全

资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

台积电利用率跟踪，预期先进节点强劲回升

台积电产能利用率跟踪

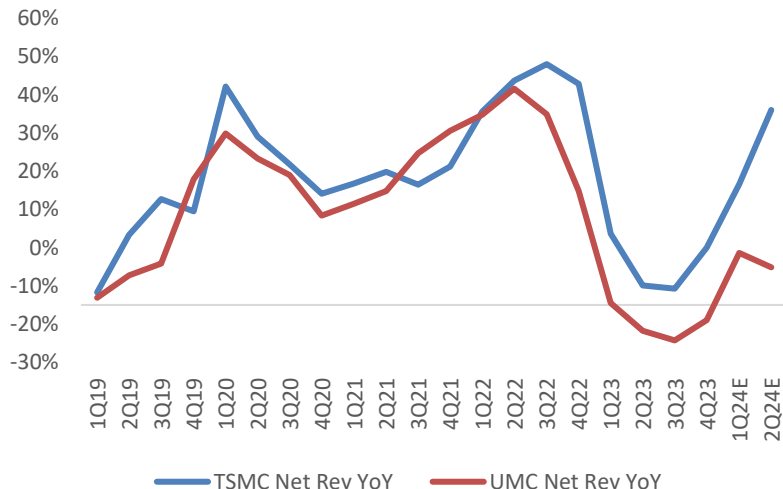
	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24E	3Q24E	4Q24E
3nm					60-65%	60-65%	70-72%	70-72%	75-80%	90-94%	97-101%	100-101%
5nm	86-90%	98-100%	100-105%	95-100%	75-80%	78-82%	88-92%	90-95%	95-98%	97-100%	99-100%	99-100%
7nm	106-108%	103-105%	95-105%	85-90%	35-40%	45-50%	58-62%	60-65%	65-70%	72-74%	75-76%	76-78%
12/16nm	106-107%	103-106%	103-110%	100-105%	78-80%	75-78%	80-85%	80-83%	70-75%	70-75%	70-75%	72-74%
22/28nm	101-104%	97-100%	105-110%	108-110%	95-100%	97-98%	92-95%	92-95%	80-82%	80-82%	82-84%	88-90%
40/45nm	100-102%	98-100%	98-100%	100-105%	85-90%	84-85%	85-88%	85-90%	84-86%	87-88%	89-90%	90-91%
55/65nm	100-102%	98-102%	98-100%	100-105%	92-94%	90-92%	85-88%	83-86%	84-86%	85-87%	85-87%	88-90%
Others	100-102%	98-100%	98-100%	98-100%	85-90%	85-90%	80-85%	78-82%	77-80%	78-80%	80-82%	82-83%

资料来源: company data, HTI

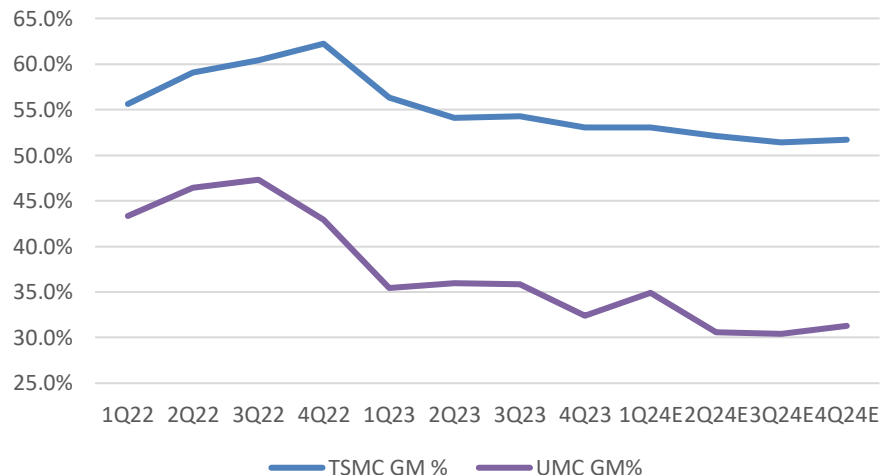
For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

同比指標觸底，一、二线强弱分明，但毛利率承压

台积电 vs. 联电 收入同比



台积电 vs. 联电 毛利率



资料来源: Gartner, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

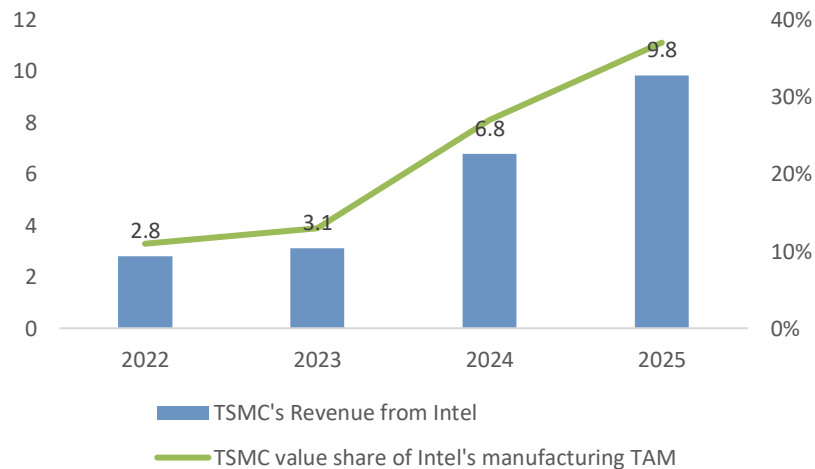
Intel 外包趋势或提速

- 我们认为英特尔将加速外包，因台积电领先的工艺和较低的成本，同时有助于英特尔的产品时间、成本结构和现金流

Intel PC 委外规划

	CPU	GPU	SoC, I/O
Meteor Lake	Intel 4	TSM N4	TSM N6
Arrow Lake H/HX	TSM N3	TSM N4	TSM N6
Lunar Lake U	TSM N3	TSM N3	TSM N3

Intel委外TAM



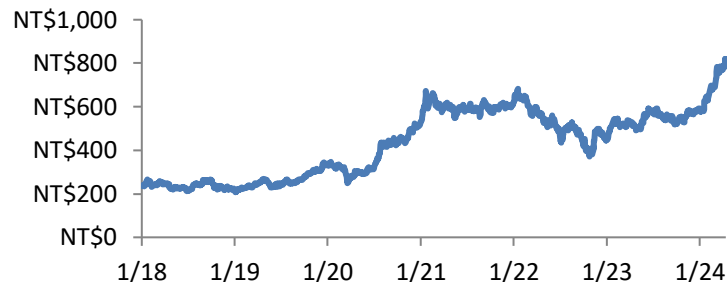
资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

台积电

行业复苏放缓无损公司增长趋势

- 由于苹果需求削减和ASML，台积电股价于近期的高点回落。尽管如此，我们认为积极因素大于消极因素，因为第二季度的指导优于市场担忧，2024年增长指引保持不变，N2开案非常强劲。长其我们仍然看好AI芯片趋势(包含edge AI)，并预计英特尔的外包将加速。
- 2024-2025年强劲的N3/N5，其次是N2：**重申我们对N3的乐观看法，系因苹果的A18 Pro（更大的芯片面积）、QCOM/MTK的N3和英特尔的外包推动的。因此，我们预计到2024年第三季度，N3 UTR将攀升至95%以上，并在2025年保持在较高水平。对于N5，在强劲的人工智能需求的推动下，我们相信UTR将在2024年保持在满载水平。展望未来，N2将是一个更强大的节点。
- 利润率预期较低，25年下半年恢复到 $\geq 53\%$ ：**由于电力/材料成本上涨、产能转换（一次性影响）和3纳米稀释时间多于预期，我们现在预计其GM在24年下半年将保持在52%以下（预期之中的）。但是，管理层表明较高的成本可能会与客户分担；加上后续UTR和规模提高，我们预计其GM将在25年下半年恢复到 $\geq 53\%$ 。



至12月31日 (NTD百万元)	2023A	2024E	2025E	2026E
营业收入	2,161,736	2,769,127	3,300,271	3,844,882
营业毛率	1,175,111	1,441,496	1,733,024	2,031,725
营业利润	921,466	1,140,309	1,376,424	1,625,125
利润总额	979,171	1,192,138	1,458,267	1,735,980
净利润	838,498	1,008,772	1,234,585	1,462,931
EPS (NTD)	32.3	38.9	47.6	56.4
获利能力 (%)				
毛利率	54.36%	52.06%	52.51%	52.84%
营业利润率	42.63%	41.18%	41.71%	42.27%
净利率	38.79%	36.43%	37.41%	38.05%
同比 (%)				
营业收入	-4.51%	28.10%	19.18%	16.50%
净利润	-17.51%	20.31%	22.39%	18.50%

- 我们预期台积电资本支出将优于预期，存储动能亦开始回复

HTI全球资本支出模型

billion USD						
	2020	2021	2022	2023	2024	2025
IDM	11	12	16	18	18	18
Memory	27	37	30	18	26	34
Foundry	27	42	49	43	42	52
World Total	65	90	95	79	87	105
Total YoY	16%	40%	5%	-17%	10%	21%

资料来源: Semi, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

ASML

订单短期递延至2季度

- 维持目标价USD1212:** 1Q24新订单为EUR3.6bn, 低于买方5-6Bn预期, 营收5.3bn也低于市场预期2.7%。尽管1Q24较弱, 我们仍预计Foundry客户会在2Q/3Q有更多订单支持2025年扩张。我们维持目前38.5bn的2025年营收和目标价
- 如何展望2Q/3Q24:** 1) 1Q24新订单不包含台积电, 正如我们preview提到台积电或将在2Q给2025年的25-30KPM 2nm产能下单。2) 我们也看到三星订单尤其是Texas 2nm/4nm晶圆厂有所延迟。目前USD6.4Bn政府补贴已经批准, 预计采购很快恢复。3) 供应链调研显示Intel仍有额外10台Hign NA订单未下, 2Q24 EUV订单或将达到历史新高。另外中国需求在高端DUV禁令之后仍然保持稳定。
- 规模和产品组合改善毛利率:** 很多投资者对ASML指引54-56%毛利率存疑, 但我们对ASML通过改善规模和产品组合实现毛利率目标有信心。一方面我们预计的80台low-NA出货 (产能90台) 和更多的High-NA出货将分摊固定成本, 另一方面NXE:3800将在2025年放量, 相比NXT:3600毛利率有高个位数提升 (价格EUR220m vs 160m) 。



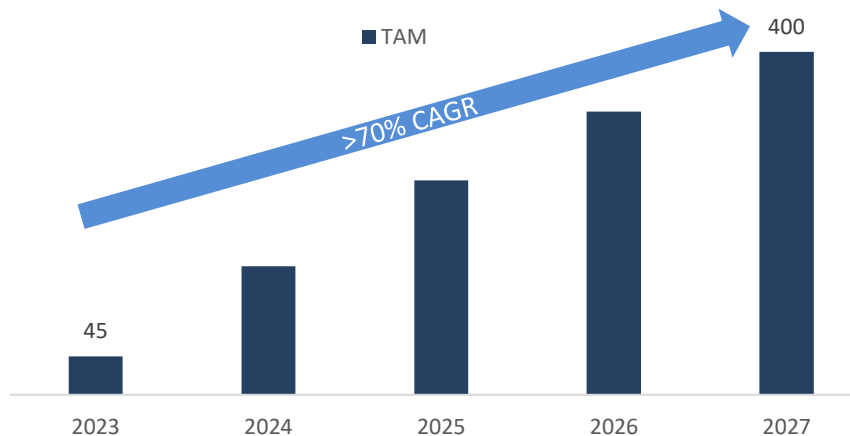
至12月31日 (百万欧元)	2022	2023	2024E	2025E
营业收入	21,173	27,559	27,754	38,577
营业毛利	10,700	14,136	14,199	20,953
营业利润	6,501	9,042	8,691	14,664
利润总额	6,456	9,084	8,739	14,789
净利润	5,624	7,839	7,515	12,621
EPS	14	20	19	32
盈利能力 (%)				
毛利率	50.54%	51.29%	51.16%	54.32%
营业利润率	30.70%	32.81%	31.32%	38.01%
净利率	26.56%	28.44%	27.08%	32.72%
YoY (%)				
营业收入	13.77%	30.16%	0.71%	39.00%
净利润	-4.40%	39.38%	-4.13%	67.93%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

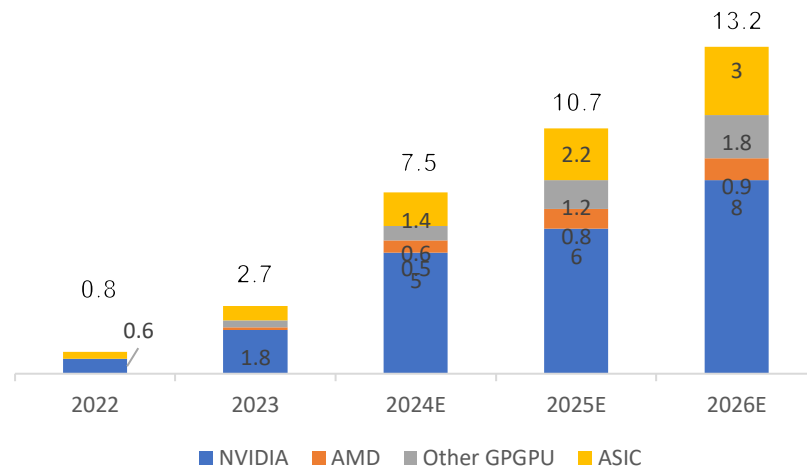
AI加速器市场规模高速增长

- AMD CEO于2023年12月AMD Advancing AI Event将2027年数据中心AI加速器TAM从\$150bn上修至\$400bn

AMD于AI加速器市场规模的看法 (USD bn)



HTI AI 芯片出货预测 (Unit m)



Source: AMD

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at [equities.htisec.com](https://www.htisec.com)

云大厂资本支出强劲，AI为主轴

北美4大 Capex (USD bn)

(USD bn)	2018	2019	2020	2021	2022	2023E	2024E
MSFT	12	14	18	23	25	35	50
Amazon	13	17	40	61	64	53	64
Google	25	24	22	25	31	32	49
Meta	14	15	15	19	31	27	38
Total Top 4 CSP	64	69	95	128	151	147	200
YoY		8%	38%	34%	19%	-3%	36%

北美4大资本支出观点梳理

	March-Q Capex	FY24 Capex Guidance
Microsoft	\$11.0bn (PP&E), +13.4% QoQ	increase YoY
Amazon	\$14.9bn, +2.3% QoQ, +5% YoY	increase YoY
Google	\$12bn, +9.1% QoQ	increase YoY
Meta	\$6.715bn, -15% QoQ	Full year CapEx to be \$35 bn ~\$40 bn

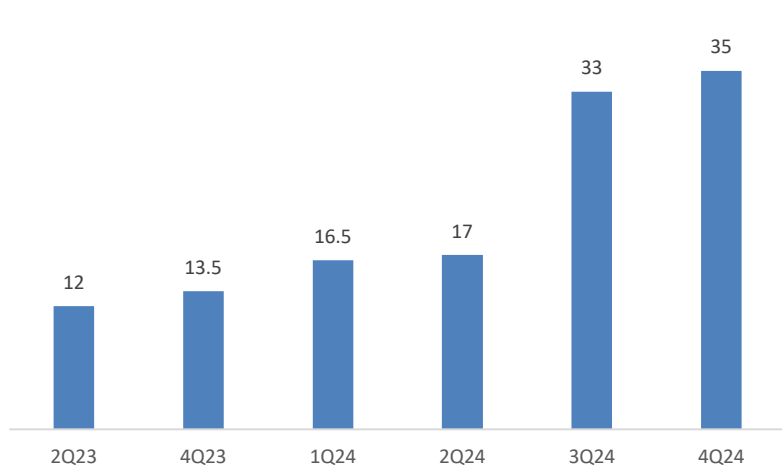
Source: Bloomberg, Company Data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

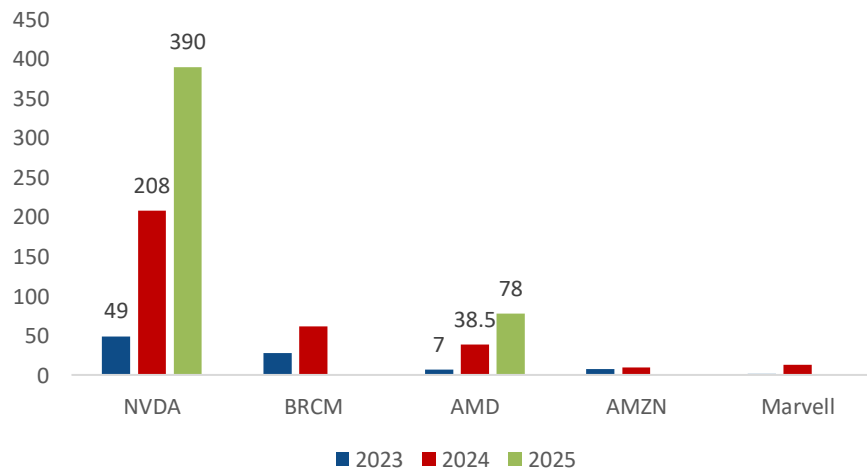
CoWoS 产能、预订量积极

- 供应链芯片订单建置积极，迭加B100/B200推出，CoWoS供需或仍紧张

台积电CoWoS产能预测 (月)



主要CoWoS客户



Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

HTI NVDA 收入模型

FY	2024	1Q25	2Q25E	3Q25E	4Q25E	2025E	1Q26E	2Q26E	3Q26E	4Q26E	2026E
CY		Apr-24	Jul-24	Oct-24	Jan-25		Apr-25	Jul-25	Oct-25	Jan-26	

Datacenter GPU (k units)

A100	507	30	-	-	-	30	-	-	-	-	-
H	899	610	733	930	800	3,073	650	300	30	-	980
B	-	-	-	85	400	485	680	950	1,150	1,140	3,920
R	-	-	-	-	-	-	-	-	-	100	110
Total	1,405	640	733	1,015	1,200	3,588	1,330	1,250	1,190	1,240	5,010
Inferencing Cards	350	515	510	200	200	1,425	250	250	200	200	900

Datacenter GPU (USD)

A100	-	10,000	-	-	-	-	-	-	-	-	-
H	-	24,065	23,500	20,833	20,833	-	20,000	20,000	20,000	-	-
B	-	-	-	31,000	32,500	-	32,500	32,800	32,000	31,500	-
R	-	-	-	-	-	-	-	-	-	40,000	-

Revenue (USD mn)

Data Center	47,503	22,563	24,272	28,502	37,042	112,379	43,294	44,627	44,759	47,067	179,748
Total revenue	60,922	26,044	28,050	32,969	41,616	128,679	47,346	48,784	49,596	52,067	197,793

Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

GPGPU路标图，Nvidia优势扩大

	MI300A	MI300X	H20	L20	H200	B100	B200	MI325X	MI350X
Vendor	AMD	AMD	Nvidia	Nvidia	Nvidia	Nvidia	Nvidia	AMD	AMD
Launch/MP	4Q23	4Q23	1Q24	1Q24	2Q24	3Q24	4Q24	4Q24	4Q25
Architecture	Chiplet	Chiplet	Monolithic	Monolithic	Monolithic	Chiplet	Chiplet	Chiplet	Chiplet
Front-end node	5nm+6nm	5nm+6nm	N4	N4	N4P	N4P	N4P	5nm+6nm	N3P
Back-end process	SoIC+CoWoS	SoIC+CoWoS	CoWoS-s	-	CoWoS-s	CoWoS-L	CoWoS-L	SoIC+CoWoS	SoIC+CoWoS
Transistor (bn)	146	153			80	208	208		
Interconnect	3D Fabric 384 GB/s (4 APU)	3D Fabric, 896 GB/s (8 GPU)	NV Link 900GB/s	PCIe4, 64GB/s	NV Link 900GB/s	NV Link, 1,800GB/s	NV Link, 1,800GB/s		
HBM (GB)	HBM3,128GB	HBM3, 192GB	96GB HBM3	48GB GDDR6	141GB HBM3E	192GB HBM3E	192GB HBM3E	288GB HBM3E	288GB HBM3E
Memory Bandwidth (TB/s)		5.2	4	0.864	4.8	8.0	8.0	6.0	
FP32	47.8 TFLOPS	47.8 TFLOPS	44 TFLOPS	59.8 TFLOPS	67 TFLOPS	-	-	-	-
FP16	980.6 TFLOPS	1.3 PFLOPS	148 TFLOPS	59.35 TFLOPS	2.0 TFLOPS	1.8 PFLOPS (Dense Core)	2.2 PFLOPS (Dense Core)		
FP4					4.0 TFLOPS	7 PFLOPS	9 PFLOPS		
TDP	600W	600W	400W	275W	700W	700W	1000W	1000W	

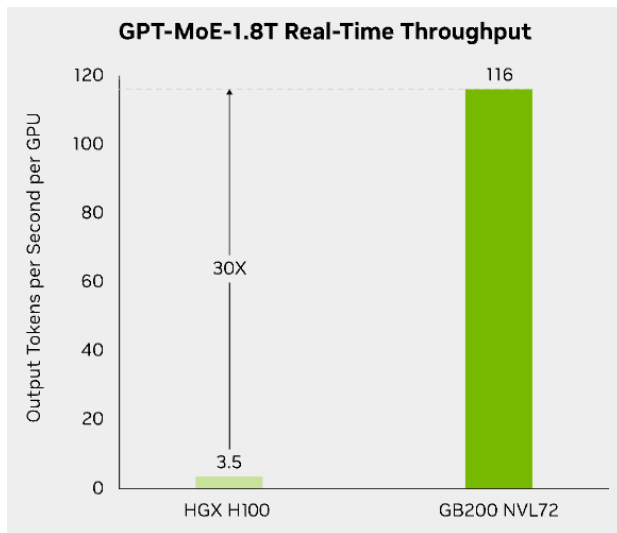
Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at [equities.htisec.com](https://www.htisec.com)

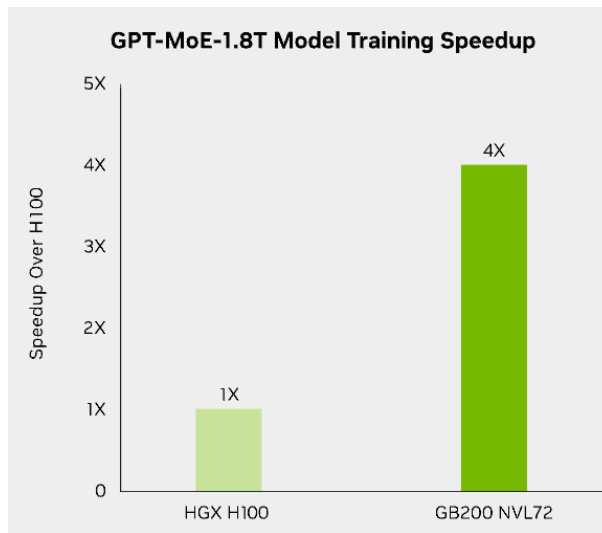
NVIDIA GB200 扩大领先差距

- GB200 NVL72相比HGX H100推理性能提升**30倍**，训练性能提升**4x**

推理性能提升30x



训练性能提升4x



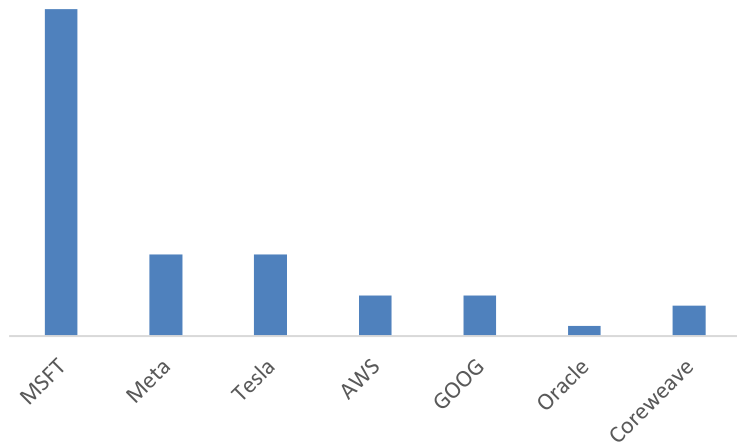
Source: NVIDIA

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

GB200 需求预测、工业富联最大份额

- 我们预期58-65k 订单 (4Q24-1Q26)，2026年的订单亦初步展开

GB200 NVL 需求分布 (k)



GB200 ODM

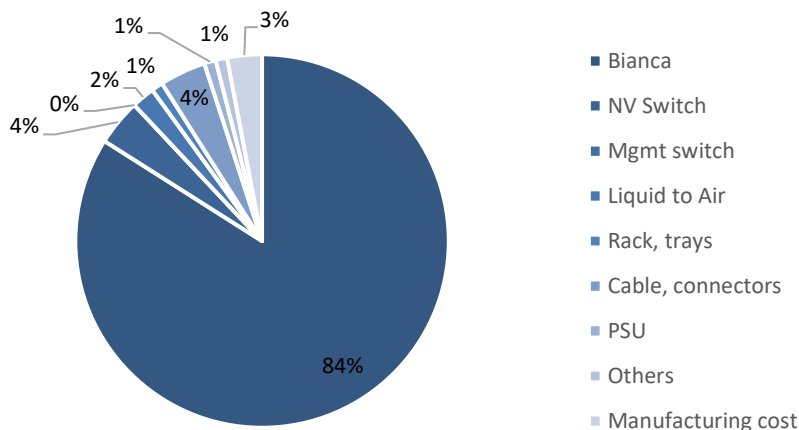
	L10	L11-L12
MSFT	富联	DELL (TBD)
Tesla		SMCI, Oracle (L10 富联)
Meta	广达	
AWS	广达	
Google	广达	
Coreweave/NV		DELL (L10 富联), SMCI

Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

GB200 NVL 聚焦利润率、鸿海集团

GB200 NVL 72 Cost



Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

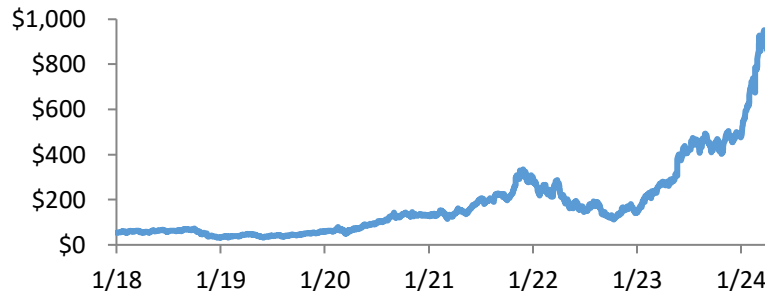
富联NVL 72利润测算、集团公司布局

(USD 000)	零组件价值	GM	GP
价值量于初期量产			
MVA			100
GB200 rack, trays	33	30%	10
GB200 Bianca module, board	27	20%	5
毛利		3.6%	115
整机单价 (1C+2G)			3,700
新增价值量于后续阶段			
In-row CDU (i.e., 8+1)	100/8	20%	2.5
Manifold	8	20%	2.0
QD	6.5	20%	1.3
预估其集团公司布局			
FIT鸿腾	内部線		
鹏鼎	PCB		

英伟达

仍然强劲

- NVDA 股价在财报前面临压力，系因一些卖方/买方担心其 25 财年第一季度的指引，然今日给出收入的指引为 240 亿美元，优于市场担忧（尽管与我们的预测完全一致），因此，我们预计将或出现股价明显反弹。尽管如此，多头和空头仍持续争论长期的持续力。就我们的观点，看好 Nvidia 向 H200/B100/B200 的快速转进，这将扩大领先优势，以及由训练模型迭代、推理、区域基础设施驱动的AI。
- H200 和 B100/B200 的快速转进：**从 H100 快速过渡到 H200/B100/B200，其中 B100/B200 的生产量将高于预期。从竞争角度来看，Nvidia 的产品将扩大领先优势，并或借力于其零组件采购优势（尤其是HBM3/3e）来限制其竞争对手（即MI300）。与此同时，ASIC 的进展仍参差不齐。
- 2025年能见度有所提高：**公司认为将受益于“通用计算到加速计算”和“生成式人工智能”的大趋势。我们长期看好：1）英伟达表示推理需求快速增长，2）多模态模型，3）区域AI建设，4）专注于GPU数据中心的CSP兴起，5）大型“to-C”公司建立LLM（i.e., 苹果）。



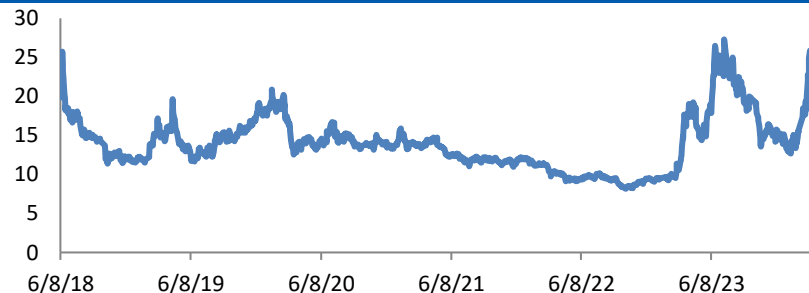
至1月31日 (USD 百万元)	2023A	2024E	2025E	2026E
营业收入	26,974	60,922	128,679	197,793
营业毛率	15,965	44,958	97,846	148,335
营业利润	9,040	37,134	84,495	125,935
利润总额	9,046	37,745	85,052	125,875
净利润	8,366	32,312	70,591	104,476
EPS (USD)	3.34	12.95	28.36	41.98
获利能力 (%)				
毛利率	59.2%	73.8%	76.0%	75.0%
营业利润率	33.5%	61.0%	65.7%	63.7%
净利率	31.0%	53.0%	54.9%	52.8%
同比 (%)				
营业收入	0.2%	125.9%	111.2%	53.7%
净利润	-25.7%	286.2%	118.5%	48.0%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

工业富联

GB200再造AI工业富联

- 工业富联近期的股价盘整显示市场对于GB200 机架方案的需求量和利润率仍持不同看法。然而，我们认为或优于市场预期。整体而言，我们仍预测公司的AI收入将在 2024 年同比增长约 1.5 倍，并在 2025 年再增长 2 倍以上。其中，GB200 NVL 36/72 是关键驱动因素。
- 重申GB200机架52k的观点：**我们指出 GB200 NVL 36/72 需求将超预期，其后市场的预期亦有所上修，但仍低于我们对58k的预期。值得注意的是，我们认为NVIDIA 的目标远超过我们的预测，係因主要客户皆已开案。对公司而言，凭借其先进优势，我们认为公司已取得了数个CSP客户，我们预计公司将在4Q24-2025年拿到GB200机架50%的份额。
- GB200 NVL 的 ODM 利润率亦佳：**最近的卖方报告预期 GB200 NVL 36/72 的毛利率为 3.5%。然而，我们认为将高于预期，因为：1) 由于其设计复杂 (特别是散热)，代工和设计费较高；2) 领先的ODMs或提供CDU、机架、模块板、tray等组件。3) 采购权已分配给ODM，代表具有降低成本的主控权。综上，我们预计公司的GB200 NVL 72毛利率4%，净利润率约为 2%，隐含 2024/25 年盈利贡献为 9 亿元/78 亿元人民币。



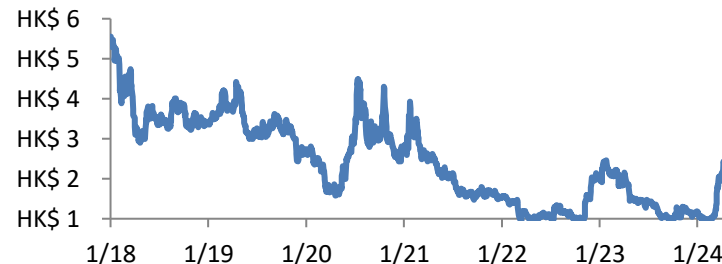
至12月31日 (人民币百万元)	2023A	2024E	2025E	2026E
营业收入	476,340	571,785	928,348	991,982
营业毛利	38,376	46,164	63,037	67,673
营业利润	22,314	27,614	40,257	43,703
利润总额	23,124	27,914	40,557	44,003
净利润	21,040	25,025	36,309	39,397
EPS(RMB)	1.06	1.26	1.83	1.98
盈利能力 (%)				
毛利率	8.1%	8.1%	6.8%	6.8%
营业利润率	4.7%	4.8%	4.3%	4.4%
净利率	4.4%	4.4%	3.9%	4.0%
YoY (%)				
营业收入	-6.9%	20.0%	62.4%	6.9%
净利润	4.8%	18.9%	45.1%	8.5%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

鸿腾精密

或进入GB200供应链

- 进入AI/GB200铜缆供应链：我们预计FIT的AI业务有望上升，因为该公司预计在2024年下半年打入GB200的铜缆，并开始供应Compute Tray内部线，随后在2025年进入switch tray。正如我们在工业富联报告中所说，我们相信ODM将会在GB200方面获得更多的采购权力；因此，凭借富士康集团的支持，我们预计FIT将在集团的供应链中占据约30%的GB200份额，约当全球市场份额为15-20%。我们预计其GB200业务的收入将在2024年达到540万美元/1.73亿美元。
- 车用、耳机支撑基本面：**EV领域连续三个季度实现三位数同比增长，主要是由于收购Voltera，我们预计这一趋势将继续。此外，FIT的System业务将受益于印度工厂在2024年底的扩产，有望从苹果下一代AirPods订单中获益，在2025年提升其组装市场份额。
- 我们预计上半年的营收将达到19.76亿美元，第二季度环比增长4.8%，毛利率同比增长2.3个百分点至18.2%。**对于现有业务，我们认为其2024年全年指引没有重大变化：1) computing部门在第二季度增长较慢，同比增长低个位数；2) system部门预计由于新的音频产品和印度工厂的扩产而实现两位数同比增长；3) 传统服务器市场逐渐恢复。



至12月31日 (USD百万元)	2023A	2024E	2025E	2026E
营业收入	4,196	4,666	5,902	6,569
营业毛利	807	943	1,238	1,403
营业利润	263	302	520	673
利润总额	179	269	500	658
净利润	129	226	414	542
EPS (USD)	0.02	0.03	0.06	0.08
获利能力 (%)				
毛利率	19.2%	20.2%	21.0%	21.4%
营业利润率	6.3%	6.5%	8.8%	10.2%
净利率	3.1%	4.8%	7.0%	8.3%
同比 (%)				
营业收入	-7.4%	11.2%	26.5%	11.3%
净利润	-24.0%	75.1%	83.5%	30.8%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

全球智能机和主要品牌出货预估 (百万台)

出货量 (百万)	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2023	2023 YoY	2024E	2024 YoY
全球智能手机	314	286	302	304	269	265	303	326	289	1163		1198	
同比	-8.9%	-8.7%	-9.7%	-17.4%	-14.5%	-7.2%	0.3%	7.4%	7.7%	-3.5%		3.0%	
OEM													
三星	74.5	63.1	64.0	59.3	60.5	53.5	59.5	53.0	60.1	227	-13%	226	0%
苹果	56.5	45.4	51.9	72.9	55.2	42.5	53.6	80.5	50.1	232	2%	217	-7%
华为	6.9	6.3	8.6	8.1	6.4	7.4	10.7	12.0	13.5	37	22%	60	64%
新荣耀	14.3	14.1	14.2	13.8	11.0	14.1	15.8	17.5	17.0	58	4%	70	20%
OPPO	41.8	37.7	39.4	36.3	35.4	35.5	37.6	38.0	35.2	147	-6%	148	1%
vivo	25.3	24.8	25.9	23.0	20.5	22.3	22.6	25.0	23.0	90	-9%	90	0%
小米	39.9	39.5	40.5	30.6	30.5	33.2	41.5	40.7	40.8	146	-3%	165	13%
市占率 %													
三星	24%	22%	21%	20%	23%	20%	20%	16%	21%	19%		19%	
苹果	18%	16%	17%	24%	21%	16%	18%	25%	17%	20%		18%	
华为	2%	2%	3%	3%	2%	3%	4%	4%	5%	3%		5%	
新荣耀	5%	5%	5%	5%	4%	5%	5%	5%	6%	5%		6%	
OPPO	13%	13%	13%	12%	13%	13%	12%	12%	12%	13%		12%	
vivo	8%	9%	9%	8%	8%	8%	7%	8%	8%	8%		8%	
小米	13%	14%	13%	10%	11%	13%	14%	12%	14%	13%		14%	

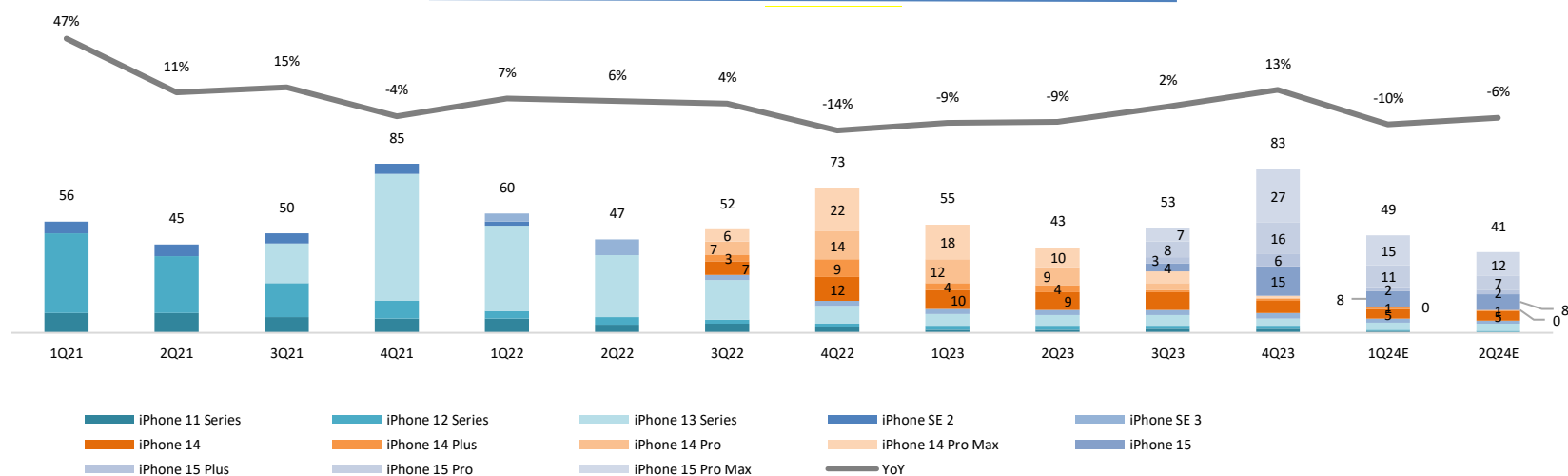
资料来源: IDC, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at [equities.htisec.com](https://www.htisec.com)

2024年于iPhone的预期已经降低

- A17 Pro/N3B表現不及预期，加上國內的高端竞争
- 2024年市场对于iPhone的预估已降低至同比下降7-10%

iPhone 组装机量预估 (百万台)



资料来源: company data, IDC, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

2024年于iPhone 16的预期已经降低

iPhone 16规格预测

	iPhone 16	iPhone 16 Plus	iPhone 16 Pro	iPhone 16 Pro Max
Launch	Sep-24	Sep-24	Sep-24	Sep-24
Display	6.1"	6.7"	6.3"	6.9"
Processor	A18, N3E	A18, N3E	A18 Pro, N3E	A18 Pro, N3E
BB	X70	X70	X75	X75
DRAM	LPDD5 8GB	LPDD5 8GB	LPDD5 8GB	LPDD5 8GB
Front camera	12MP, 6P	12MP, 6P	12MP, 6P	12MP, 6P
Rear camera	48MP 7P, 12MP 5P	48MP 7P, 12MP 5P	48MP 7P, Periscope 12MP 1G+3P, 48MP 6P	48MP 7P, Periscope 12MP 1G+3P 48MP 6P
Face ID	Structured light	Structured light	Structured light	Structured light
Casing	Aluminium	Aluminium	Titanium	Titanium
Others	a new capture button	a new capture button	a new capture button	a new capture button

资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

今年为产品过渡期

主要产品线组装量 (mn units)

(mn)	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	2022	2023	2024
iPhone	52.0	73.0	54.5	43.0	53.0	81.0	232	233	218
iPad	13.5	20.5	12.0	9.0	12.5	14.0	59	48	52
Watch	10.0	14.5	8.5	7.5	11.5	12.5	43	40	39
Mac	10.1	7.5	4.1	5.3	7.2	5.7	30	22	22
AirPods	17.1	21.1	14.1	12.1	15.1	16.1	72	57	55
YoY									
iPhone	4%	-14%	-9%	-9%	2%	11%	-1%	0%	-7%
iPad	-10%	32%	0%	-31%	-7%	-32%	3%	-19%	8%
Watch	0%	0%	-15%	-12%	15%	-14%	6%	-7%	-3%
Mac	61%	-17%	-23%	0%	-35%	-100%	2%	-25%	0%
AirPods	-12%	-22%	-26%	-20%	-12%	-24%	-7%	-21%	-4%

资料来源: company data, IDC, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

Apple AI 陆续释出，持续观察外部合作

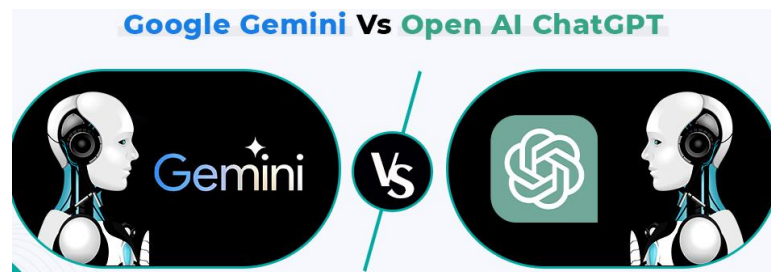
M4芯片具备端测模型运算能力



手机芯片尺寸放大

	A17 Pro	A18 Pro	A19 Pro
Process Node	N3B	N3E	N3P
CPU cores	6 cores, 10% better than A16	Die area +8-10%	Die shrink 5% Die area ?
GPU cores	6 cores, 20% faster than A16		
NPU cores	16 cores, x2 faster than A16		
iPhone models	iPhone 15 Pro	iPhone 16 Pro	iPhone 17 Pro

苹果AI今年关注外部合作



Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

2025-2026或将重启强劲产品周期

iPhone 17规格预测

	iPhone 17	iPhone 17 Slim	iPhone 17 Pro	iPhone 17 Pro Max
Launch	Sep-25	Sep-25	Sep-25	Sep-25
Display	6.1"	6.6"	6.3"	6.9"
Processor	A18, N3E or A19 N3P	A18, N3E or A19 N3P	A19 Pro, N3P	A19 Pro, N3P
DRAM	LPDD5 8GB	LPDD5 8GB	LPDD5 12GB	LPDD5 12GB
Front camera	24MP, 6P	24MP, 6P	24MP, 6P	24MP, 6P
Rear camera	48MP 7P, 12MP 5P	48MP 7P, 12MP 5P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P
Face ID	Structured light	Structured light	Structured light	Structured light, Metalens
Casing	Aluminum (a more complex design)	Aluminum (a more complex design)	Aluminum (a more complex design)	Titanium

折叠装置预测

	Foldable device	Foldable device
MP	Late-2025	Late-2026
Display	20.3"	7.9" or 8.3"

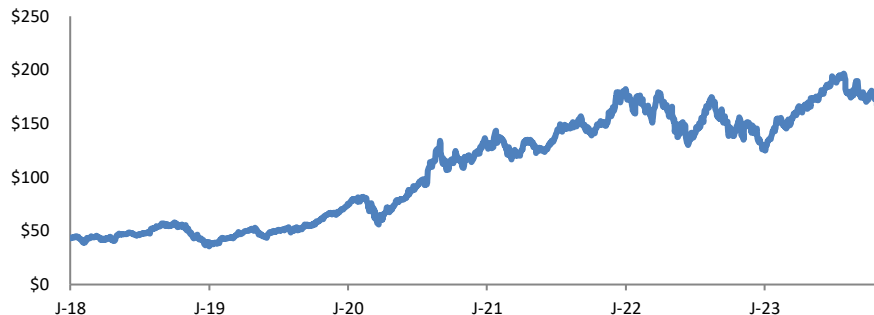
资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

Apple

强劲产品周期于明年启动

- 苹果股价自去年首次覆盖以来的表现低于大盘，观察近日市场预期，我们认为 iPhone 15 的弱势已经反映，且市场对于 2024 年整年的预期较低，因此判断下行风险有限。后续市场将聚焦AI和明年之后的产品周期，我们当前认为公司将自明年重启强劲的产品周期，驱动力为AI、iPhone 17和折叠装置。此外，苹果于 5 月 3 日表示，公司将在未来几周内释出更多人工智能的信息，亦有利于股价。
- 2024 年的市场预期低，竞争的影响或将收窄：**市场的预期业已降低，我们预计 iPhone 同比将在 2H24 有所改善，系因国内竞争的同比影响将减轻，且下半年 SoC的差距将较先前预期大
- iPhone 17、折叠机将重拾产品力：**我们预期iPhone 17 (2025 年) 或为一个强劲的周期，系因更完备的AI功能，和硬件上新的外观/外壳、相机 (48MP 潜望镜、24MP 前制)、AI 处理能力 (A19 Pro、12GB DRAM)、更小的 Dynamic Island (Pro Max) 和新的轻薄Slim机种 (取代 Plus)。其中AI功能将需要通过更强的 芯片和内存来支持，这将是换机的主要动力之一。此外，我们认为苹果将在 2025 年末量产 20.3 英寸折叠平板，并已开案并规画于 2026 年末量产折叠手机，此将进一步提升产品力



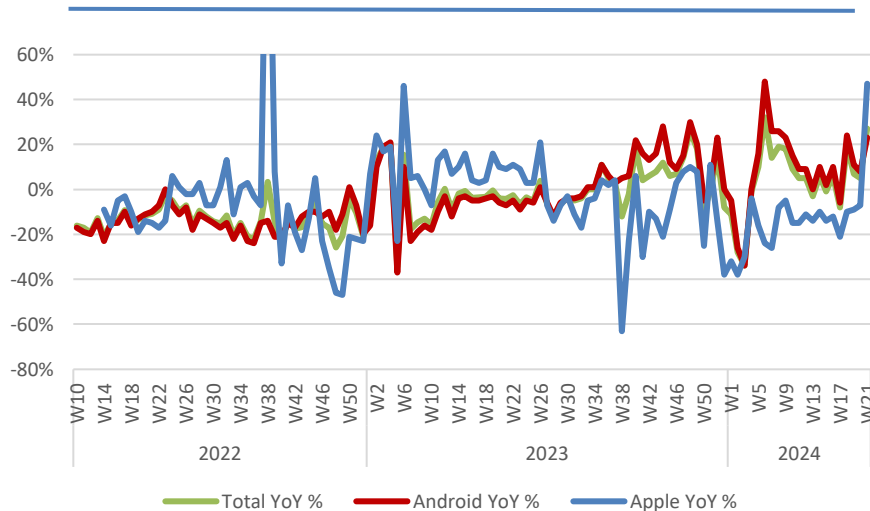
至1月31日 (USD百万元)	2023A	2024E	2025E	2026E
营业收入	383,285	387,666	407,606	440,253
营业毛利率	169,148	178,499	189,372	207,518
营业利润	114,301	120,446	129,472	143,418
利润总额	113,736	120,654	129,672	143,618
净利润	96,995	101,461	108,925	120,639
EPS (USD)	6.13	6.59	7.34	8.38
获利能力 (%)				
毛利率	44.13%	46.04%	46.46%	47.14%
营业利润率	29.82%	31.07%	31.76%	32.58%
净利率	25.31%	26.17%	26.72%	27.40%
同比 (%)				
营业收入	-2.80%	1.14%	5.14%	8.01%
净利润	-2.81%	-4.60%	-7.4%	-10.8%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

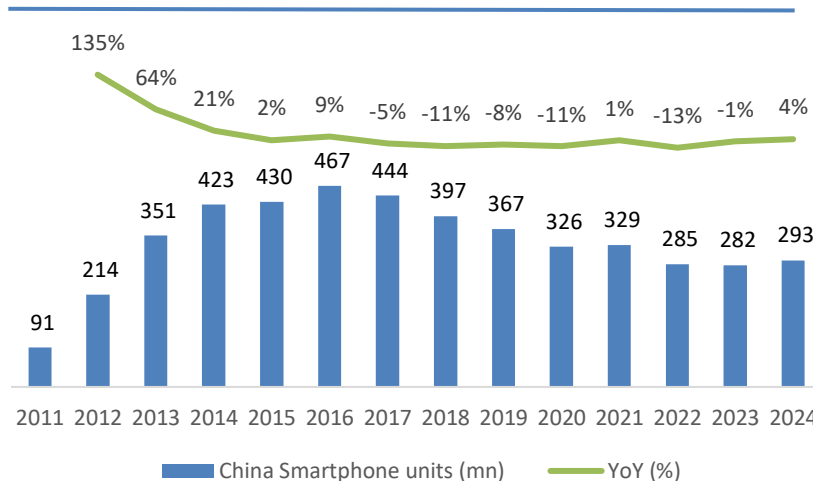
国内手机需求走平，苹果份额回升

- 去年12月后拉货放缓，需求偏弱，库存调整持续
- 5月中后618促销启动，同比回升，苹果受益降价而使得份额回升

大陆智能机市场销量同比



国内智能手机市场预期



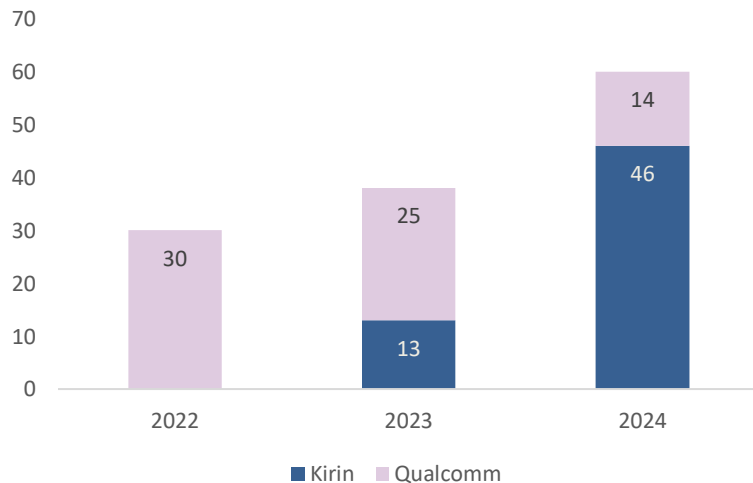
资料来源: BCI, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

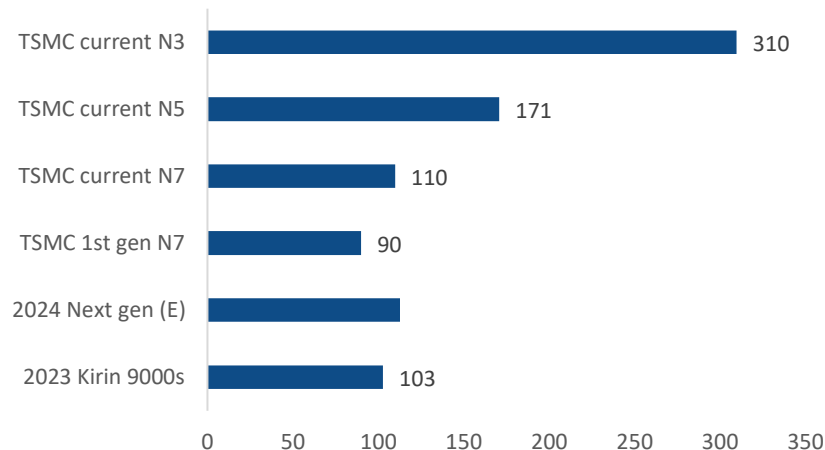
H手机为主要市场增量，然动能趋缓

- Mate 60 使苹果承压，然中阶Nova 12放量低于预期
- 主芯片预计节点推进较慢，下半年动能待观察

H智能机出货&组合预估 (mn units)



晶体管密度路标图 (mtr/mm²)



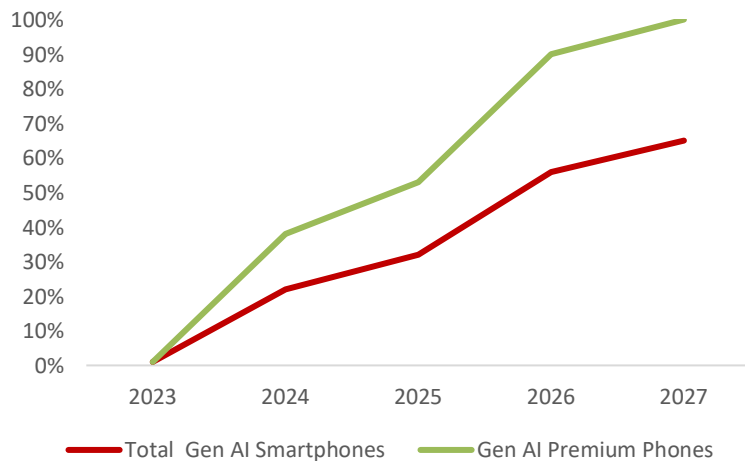
资料来源: IDC, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

AI手机，芯片首先受益

- Gartner定义AI手机为具有端上运算模型的能力，预期AI芯片将加速推出

Gartner预估AI手机渗透率



AI进一步推动芯片单价上升

	2H22	2H23	2H24
165-180			8 Gen 4
145-165		8 Gen 3	D9400
130-140	8 Gen 2		
130		D9300	
120	D9200		
55-65		D8300	D8400

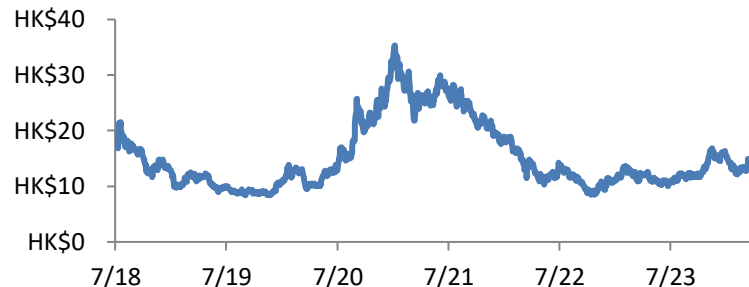
Source: Gartner, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

小米集团

季报前瞻，增长加速

- 1季报前瞻:** 小米预计将于5月下旬举行财报会议，我们预测其调整后净利为55亿，同比增长71%，主要驱动力为：1) 智能手机毛利率高于预期，2) IoT毛利率高于预期，3) 费用控制。由于业绩系由本业推动，预期将为一高质量财报
- SU7的购买等待时间延长至约8个月:** 4月份SU7的交付量约为7k，已锁单的量达8.8万辆，对应年产能为10万辆（至6月），这表明等待时间将持续延长(=股价的关键催化剂)。再加上我们对2025年上半年新SUV车型的预期，我们预测2024/2025年的出货量为10万/20万辆。此外，管理层预计毛利率为+5-10%，然我们预测将处于上缘，归功于单车款的规模和较低的零件成本。
- 智能手机业务稳定且有韧性:** 在新兴市场份额增长和1季度出货量强劲的推动下，我们预计其2024年智能手机出货量将同比增长14%。我们还预计其在中国高端市场的份额将稳定向上，下半年将得益于较早推出3nm机种。此外，由于产品组合、零部件本地化和SKU减少，我们预计其毛利率将在较高水平。



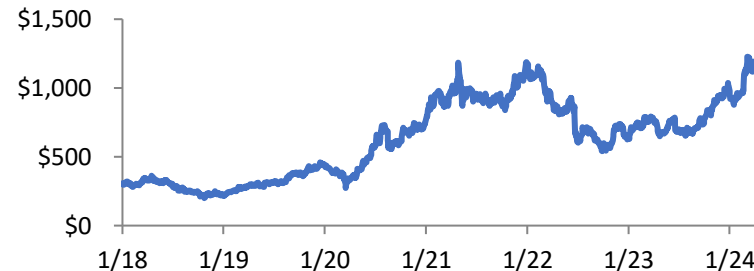
至12月31日 (人民币百万元)	2022	2023E	2024E	2025E
营业收入	280,044	270,970	324,130	361,291
营业毛利率	47,577	57,476	63,859	67,122
营业利润	2,816	20,009	18,389	18,272
利润总额	9,949	23,810	22,563	22,692
净利润	8,489	19,260	18,510	18,614
EPS (人民币)	0.35	0.79	0.76	0.77
获利能力 (%)				
毛利率	17.0%	21.2%	19.7%	18.6%
营业利润率	1.0%	7.4%	5.7%	5.1%
净利率	3.0%	7.1%	5.7%	5.2%
同比 (%)				
营业收入	-89%	610%	-8%	-1%
净利润	-62%	127%	-4%	1%

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

联发科技

指引较预期佳

- 整体财报会议比预期佳**，系因2季度指引较市场担忧为佳，加上2024 年收入/利润率指引维持不变，然而缺乏特别股利的指引唯一负面因子。短期来看，我们认为智能手机的调整是小幅调整（但能见度低），而高端需求仍持稳。也就是说，公司的AI 旗舰芯片（即 D9300/D9400/D8300）将持续推动更佳的产品组合，並搭上边缘 AI 趋势；另外加上电视、WiFi 和 PMIC 业务企稳，我们预其公司2024/2025 年收入同比增长约 20%/17%。
- 指引好于预期**：1季度的业绩表现强劲，收入1,334.58亿元，季增3%，年增40%，高于预期，受益于手机芯片补货、平均售价上升，电视和宽带业务回升。针对第2季的指引，以美元汇率32元计算，收入与前季相比持平至下滑9%，或年增24至26%，中间值-4.5% QoQ优于市场预期的高个位数下降。毛利率指引在47%，维持高位。2024全年收入指引维持先前预期mid-teen 增长。



至12月31日 (NTD百万元)	2022A	2023	2024E	2025E
营业收入	548,796	433,446	531,992	621,769
营业毛率	270,904	207,367	258,396	294,737
营业利润	126,788	71,800	105,254	117,603
利润总额	135,561	86,782	113,961	123,757
净利润	118,141	76,979	98,935	107,010
EPS (NTD)	74.4	48.5	62.3	67.4
获利能力 (%)				
毛利率	49.4%	47.8%	48.6%	47.4%
营业利润率	23.1%	16.6%	19.8%	18.9%
净利率	21.5%	17.8%	18.6%	17.2%
同比 (%)				
营业收入	11.2%	-21.0%	22.7%	16.9%
净利润	6.0%	-34.8%	28.5%	8.2%

美国持续出台AI限制

- 2022年10月美国出台BIS限制法案禁止向中国出口高性能芯片（如A/H100），2023年进一步增加限制（A/H800, L40S）
- 海外芯片厂商后续或无法进一步迭代产品

BIS限制法案演变

时间	2022年10月7日	2023年10月17日
限制内容	算力 > 4800 TOPS且带宽 > 600GB/S	3A090.a 1)TPP>4800 2)TPP > 1600且PD>5.92 3A090.b 1) 2400 < TPP < 4800 2) TPP > 1600, 且 3.2 < PD < 5.92
		将先进芯片的出口许可要求扩大到美国维持武器禁运的所有22个国家以及中国澳门
		将壁仞科技、摩尔线程涉及先进计算芯片开发的中国实体及子公司(13家实体)添加到实体清单
影响产品	NVIDIA A/H100	NVIDIA A/H800, L40S AMD MI300

Source: BIS, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at [equities.htisec.com](https://www.htisec.com)

国内算力中心利好政策

- 国内各地政府出台大量利好政策支持AI服务器智算中心发展

《算力基础设施高质量发展行动计划》

	序号	指标	2023	2024	2025
计算力	1	算力规模 (EFLOPS)	220	260	300
	2	智能计算中心 (个)	30	40	50
	3	智能算力占比 (%)	25	30	35
运载力	4	重点应用场所光传送网 (OTN) 覆盖率 (%)	50	65	80
	5	SRv6 等创新技术使用占比 (%)	20	30	40
	6	国家枢纽节点数据中心 集群间网络时延达标率 (%)	65	75	80
存储力	7	存储总量 (EB)	1200	1500	1800
	8	先进存储容量占比 (%)	25	28	30

各地政府智算中心政策

部门	政策名称	重点内容
宁夏回族自治区人民政府办公厅	《自治区人民政府办公厅关于促进全国一体化算力网络国家枢纽节点宁夏枢纽建设若干政策的意见》	加大信创企业扶持力度，对基础软硬件实现 国产化率90% 以上的数据中心，给予企业最高不超过1000万元奖励。
宁夏自治区发展改革委	《全国一体化算力网络国家枢纽节点宁夏枢纽建设2023年工作要点》	推动龙心中科、中电子、中兴、华为等自主可控产业园区落地，鼓励以国产化CPU、GPU、操作系统等自主可控产品为底座的信创云平台自主研发，打造安全可信计算、网络和存储能力
青岛人工智能产业园	《青岛“海之心”人工智能计算中心项目》	要求 国产人工智能加速卡占比>70%
成都市经信局	《成都市围绕超算智算加快算力产业发展的政策措施》	鼓励智算中心建设 国产自主可控、安全可靠 的人工智能算力基础设施和技术路线生态
南京麒麟科创园	《南京智能计算中心算力推广办法（试行）》	以“算力券”和“算力折扣”两种形式为广大企业提供更优质的算力资源、更有力的服务保障
北京市政府	《北京市加快建设具有全球影响力的人工智能创新策源地实施方案（2023-2025年）》	在人工智能产业聚集区新建或改建升级人工智能商业化算力中心，加强 国产芯片 部署应用，推动自主可控软硬件算力生态建设
成都市经信局	《成都市加快大模型创新应用推进人工智能产业高质量发展的若干措施》	引导国家超算成都中心、成都智算中心合理扩容， 支持鲲鹏、昇腾、海光等自主可控芯片部署 ，提高自主研发算力设备比例

Source: Government, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at [equities.htisec.com](https://www.htisec.com)

本土AI投资快速增长

- 互联网厂商资本支出于2024年大幅成长，AI或为主要推手
- 中国10亿参数规模以上的大模型数量已超100个

中国互联网厂商资本支出

RMB bn	2018	2019	2020	2021	2022	2023	2024E	2025E
腾讯	22.2	27.1	39.4	31.0	23.2	21.4	46.4	47.3
百度	8.8	6.4	5.1	10.9	8.3	11.2	12.9	13.1
阿里	49.6	45.4	43.2	53.3	34.3	32.1	42.3	45.3

Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

本土AI投资快速增长

- 政府智算中心项目以及运营商采购为国内AI算力另一重要来源
- 已经宣布的政府智算项目投资额或超过180亿元

中国运营商AI算力服务器2023-24集中采购

	总金额 (亿)	AI服务器 (台)	平均单价 (万)	中标人	份额/亿
中国电信	85	4175	204	超聚变	17.9
				浪潮	12.8
				新华三	12.4
				华鲲振宇	8.6
				烽火	7.1
				昆仑技术	6.1
				其他	20.1
中国移动	191	7994	239	昆仑技术	40.2
				华鲲振宇	33.5
				宝德	30.2
				百信	26.8
				武汉长江	23.1
				神州鲲泰	20.1
				湘江鲲鹏	16.8
中国联通	20	2503	80	昆仑技术	
				宝德	
				四川虹信	
				武汉长江	

Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

H公司领衔国产芯片进展

- 昇腾910B性能单卡性能接近A100，小规模集群较A100仍有差距，但大规模集群性能进一步下降
- 除性能外互联能力，软件生态也是重要考量因素
- 后续新品性能进一步提升；机柜方案同样值得关注

主要算力芯片性能对比

Company	Product	Computing power@INT8	FP16 performance	Architechure	Memory bandwidth
Nvidia	B200	4,500 TOPS	2,250 TFLOPS	Blackwell	8TB/s
Nvidia	H100	1,980 TOPS	990 TFLOPS	Hopper	3.35TB/s
Nvidia	A100	624 TOPS	312 TFLOPS	Ampere	1935 GB/s
HiSilicon	昇腾910	640 TOPS	320 TFLOPS	Da Vinci	
HiSilicon	昇腾310	16 TOPS	8 TOPS	Da Vinci	
Hygon	深算一号				1024GB/s
Cambricon	MLU370	256 TOPS	96 TFLOPS	LUarch03	
AMD	MI300X	2,600 TOPS	1,300 TFLOPS	CDNA3	5.3TB/s

Source: Company, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

AI服务器供应链

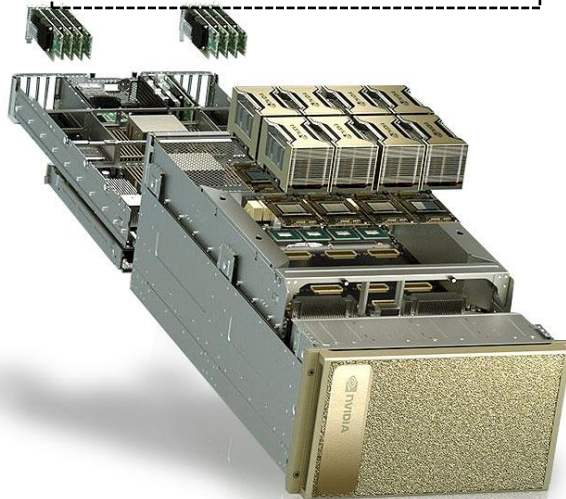
- GPU
- 本土：华为海思、海光信息、寒武纪、景嘉微
- 海外：Nvidia、AMD、INTC

- CPU
- 本土：华为海思、海光信息、飞腾信息、龙芯中科
- 海外：INTC、AMD、Nvidia

- 电源模块
- 本土：中国长城、欧陆通、杰华特、晶丰明源
- 海外：MPWR、英飞凌

- PCB/CCL
- 本土：沪电股份、胜宏科技、深南电路、生益电子、生益科技、华正新材
- 台湾/海外：欣兴、金像电、TTM、台光电、台耀

- DRAM
- 本土：长鑫存储、昇维旭、福建晋华、澜起科技
- 海外：三星、美光、海力士



- 连接器、铜缆
- 本土：华丰科技、鼎通科技、方邦股份
- 海外：安费诺

- ODM/OEM
- 本土：华鲲振宇、拓维信息、神州数码、烽火通讯等
- 台湾：广达、纬创、英业达

- 散热
- 本土：英维克、同飞股份
- 台湾/海外：双鸿、AVC、VRT

- NAND/SSD
- 本土：长江存储、兆易创新、金泰克、佰维存储
- 海外：三星、美光、海力士

- IC载板
- 本土：兴森科技、深南电路、
- 台湾/海外：欣兴、Ibiden、南电

- 交换机
- 本土：菲菱科思、共进股份
- 海外/台湾：Arista、智邦、CLS

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

短期供给仍有瓶颈

- 短期产能仍然限制国内AI芯片、AI算力芯片为最高生产优先级
- 后续AI新产品将进一步增大产能消耗
- 后续关注国内头部晶圆厂扩产进度

主要先进制程芯片

先进制程芯片	良率	产能需求	优先级
昇腾系列	低	高	高
鲲鹏系列	低	较高	较高
Kirin系列	较高	高	高
车载MDC810	较高	较低	较低
FPGA	高	低	低

资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

大基金三期为重要投资背景

- 国家大基金三期正式成立，注册资本达3440亿元（第一期规模987.2亿元、第二期2014.5亿元）
- 大基金一期重点明确-半导体制造：中芯南方、中芯北方、中芯集成、华虹、长江存储、长电科技、通富微电等、半导体设备龙头北方华创、国产EDA龙头华大九天等均在列
- 大基金二期于半导体设备与材料的力度明显加大，共投资7家半导体材料企业和6家半导体设备企业
- **大基金三期投资重点？**
 - 先进工艺制造（逻辑、存储）和先进封装
 - 关键半导体设备、材料、零部件，比如光刻机产业链、光刻胶
 - AI算力芯片及HBM

资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

• 大陆厂商逐步实现“0到1”的突破

光刻机组件及配套设施	公司	股票代码	目前进展
物镜系统	国望光学	-	研发光刻机曝光光学系统
	赛微电子	300456.SZ	给ASML提供透镜系统MEMS部件和晶圆制造服务
光源	科益虹源	-	研发准分子激光器
	福晶科技	002222.SZ	研发KBBF晶体(用于激光设备的上游关键零部件), KBBF晶体是目前可直接倍频产生EUV激光的非线性光学晶体
光学元件	茂莱光学	688502.SH	光刻机光学透镜提供商, 目前生产的半导体光学透镜被应用在光刻机光学系统中
	腾景科技	688195.SH	公司多波段合分束器已完成产品开发, 进入半导体设备厂供应链
	炬光科技	688167.SH	为上海微电子等企业提供了半导体激光退火系统以及核心元器件
	晶方科技	603005.SH	子公司Anteryon为全球领先的光学设计和晶圆级光学镜头制造商, 是ASML光学平台和晶圆对位传感器供应商
	苏大维格	300331.SZ	向上海微电子提供了光刻机用的定位光栅产品
浸没式系统	启尔机电	-	浸没系统供应商
双工作台	华卓精科	A20224.SH	双工件台供应商
空气净化设备	美埃科技	688376.SH	为上海微提供了光刻设备所需的洁净过滤产品
结构零部件	富创精密	688409.SH	半导体零部件制造的工艺技术达到主流国际客户标准, 是ASML的战略供应商
	新莱应材	300260.SZ	真空产品以及气体产品均可以应用到光刻机的设备中

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

半导体材料替代空间广阔

- 光刻胶/靶材/掩模版/可剥铜/ABF膜代替空间大

半导体材料市场空间及竞争格局

	2022年国内市场空间 (亿美元)	日本厂商	日商市占率	国内厂商	2022年国产化率
半导体材料					
硅片	30	Sumco, 信越	50%	沪硅产业, 中环股份, 立昂微	30%
光刻胶	9.7	富士胶片, JSR, 东京应化, 信越化学	70%	彤程新材, 华懋科技, 南大光电, 容大感光, 永太科技	ArF<1%, KrF~5%
靶材	1.6	日矿金属, 东曹	50%	江丰电子, 阿石创, 有研新材	~40%
电子特气	10.5	太阳日酸	18%	华特气体, 金宏气体, 雅克科技	15%
CMP	5.7	日立, Fujimi	26%	安集科技, 鼎龙股份	37%
掩模版	9.7	DNP, Toppan	50%	清溢光电, 路维光电	4%
湿化学品	4.9	Kanto, 东京应化, UBE, 住友化学, 三菱化学等	29%	江化微, 上海新阳	25%~30%
可剥铜	4	三井金属	80%	方邦股份	<1%
ABF膜	3.2	味之素、太阳油墨	>98%	华正新材、生益科技	<1%
ABF载板	7.2	Ibiden, Shinko, Kyocera	25%	兴森科技, 深南电路	<5%
引线框架	6.8	三井高科技, Newkon	21%	康强电子	11%

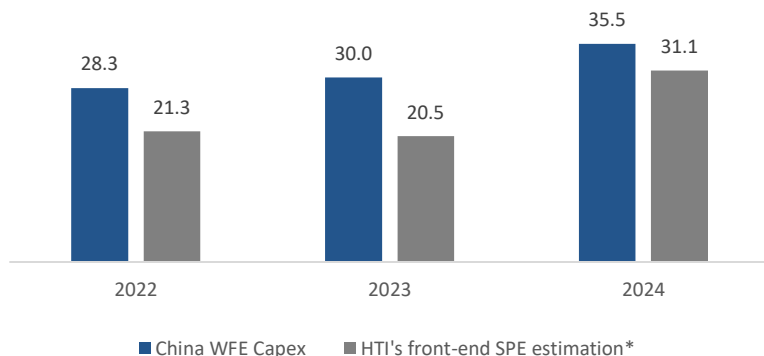
资料来源: Semi, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

先进制程带动国内设备投资周期走强

- 竞争格局无需过度担忧
- 先进制程将驱动增长，我们预估2025年先进存储以及逻辑将占本土WFE支出超40%

国内WFE市场预估 (USD bn)



国内主要扩产项目 (KPM)

Company Name	Fab name	Location	Tech	2024E	2025E
PXW/PXX		Shenzhen	12" mature	20	20
SMIC	SMBC	Beijing	12" mature	40	30
SMIC		Tianjin	12" mature	10	40
SMIC		Shanghai	12" mature	30	40
HH Grace	Fab 9	Wuxi	12" mature	0	30
Nexchip	N3/N4	Hefei	12" mature	40	50
Cansemi		Guangzhou	12" mature	20	20
YMTC	Fab3	Wuhan	3D NAND	30	40
CXMT	Fab-2/Fab-3	Hefei	DRAM	80	70
Swaysure		Shenzhen	DRAM		

资料来源: Company data, HTI

For full disclosure of risks, valuation methodologies and target price formation on all HTI rated stocks, please refer to the latest full report on our website at equities.htisec.com

IMPORTANT DISCLOSURES

This research report is distributed by Haitong International, a global brand name for the equity research teams of Haitong International Research Limited (“HTIRL”), Haitong Securities India Private Limited (“HSIPL”), Haitong International (Japan) K.K. (“HTIJKK”), Haitong International Securities Company Limited (“HTISCL”), and any other members within the Haitong International Securities Group of Companies (“HTISG”), each authorized to engage in securities activities in its respective jurisdiction.

Analyst Certification:

I, Jeff Pu, certify that (i) the views expressed in this research report accurately reflect my personal views about any or all of the subject companies or issuers referred to in this research and (ii) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report; and that I (including members of my household) have no financial interest in the security or securities of the subject companies discussed.

Conflict of Interest Disclosures

HTI and some of its affiliates may engage in investment banking and / or serve as a market maker or hold proprietary trading positions of certain stocks or companies in this research report. As far as this research report is concerned, the following are the disclosure matters related to such relationship (As the following disclosure does not ensure timeliness and completeness, please send an email to ERD-Disclosure@htisec.com if timely and comprehensive information is needed).

IMPORTANT DISCLAIMER

For research reports on non-Indian securities: The research report is issued by Haitong International Research Limited (“HTIRL”), a wholly owned subsidiary of Haitong International Securities Group Limited (“HTISGL”) and a licensed corporation to carry on Type 4 regulated activity (advising on securities) for the purpose of the Securities and Futures Ordinance (Cap. 571) of Hong Kong, with the assistance of Haitong International (Japan) K.K. (“HTIJKK”), a wholly owned subsidiary of HTISGL and which is regulated as an Investment Adviser by the Kanto Finance Bureau of Japan.

For research reports on Indian securities: The research report is issued by Haitong Securities India Private Limited (“HSIPL”), an Indian company and a Securities and Exchange Board of India (“SEBI”) registered Stock Broker, Merchant Banker and Research Analyst that, inter alia, produces and distributes research reports covering listed entities on the BSE Limited (“BSE”) and the National Stock Exchange of India Limited (“NSE”) (collectively referred to as “Indian Exchanges”). HSIPL was acquired and became part of the Haitong International Securities Group of Companies (“HTISG”) on 22 December 2016.

All the research reports are globally branded under the name Haitong International and approved for distribution by Haitong International Securities Company Limited (“HTISCL”) and/or any other members within HTISG in their respective jurisdictions.

The information and opinions contained in this research report have been compiled or arrived at from sources believed to be reliable and in good faith but no representation or warranty, express or implied, is made by HTIRL, HTISCL, HSIPL, HTIJKK or any other members within HTISG from which this research report may be received, as to their accuracy, completeness or correctness. All opinions expressed herein are as of the date of this research report and are subject to change without notice. This research report is for information purpose only. Descriptions of any companies or their securities mentioned herein are not intended to be complete and this research report is not, and should not be construed expressly or impliedly as, an offer to buy or sell securities. The securities referred to in this research report may not be eligible for purchase or sale in some jurisdictions. If an investment product is denominated in a currency other than an investor's home currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Certain transactions, including those involving derivatives, give rise to substantial risk and are not suitable for all investors. You should also bear in mind that recommendations in this research report are not tailor-made for you. The analyst has not taken into account your unique financial circumstances, such as your financial situation and risk appetite. You must, therefore, analyze and should, where applicable, consult your own legal, tax, accounting, financial and other professional advisers to evaluate whether the recommendations suits you before investment. Neither HTISG nor any of its directors, employees or agents accepts any liability whatsoever for any direct or consequential loss arising from any use of the materials contained in this research report.

HTISG and our affiliates, officers, directors, and employees, excluding the analysts responsible for the content of this document, will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research report. Sales, traders, and other professionals of HTISG may provide oral or written market commentary or trading strategies to the relevant clients and the companies within HTISG that reflect opinions that are contrary to the opinions expressed in this research report. HTISG may make investment decisions that are inconsistent with the recommendations or views expressed in this research report. HTI is under no obligation to ensure that such other trading decisions, ideas or recommendations are brought to the attention of any recipient of this research report.

Please refer to HTI's website www.equities.htisec.com for further information on HTI's organizational and administrative arrangements set up for the prevention and avoidance of conflicts of interest with respect to Research.

Non U.S. Analyst Disclosure: The HTI analyst(s) listed on the cover of this Research is (are) not registered or qualified as a research analyst with FINRA and are not subject to U.S. FINRA Rule 2241 restrictions on communications with companies that are the subject of the Research; public appearances; and trading securities by a research analyst.

DISTRIBUTION AND REGIONAL NOTICES

Except as otherwise indicated below, any Recipient wishing to discuss this research report or effect any transaction in any security discussed in HTI's research should contact the Haitong International salesperson in their own country or region.

Notice to Hong Kong investors: The research report is distributed by Haitong International Securities Company Limited ("HTISCL"), which is a licensed corporation to carry on Type 1 regulated activity (dealing in securities) in Hong Kong. This research report does not constitute a solicitation or an offer of securities or an invitation to the public within the meaning of the SFO. This research report is only to be circulated to "Professional Investors" as defined in the SFO. This research report has not been reviewed by the Securities and Futures Commission. You should not make investment decisions solely on the basis of the information contained in this research report. Recipients of this research report are to contact HTISCL salespersons in respect of any matters arising from, or in connection with, the research report.

Notice to U.S. investors: As described above, this research report was prepared by HTIRL, HSIPL or HTIJKK. Neither HTIRL, HSIPL, HTIJKK, nor any of the non U.S. HTISG affiliates is registered in the United States and, therefore, is not subject to U.S. rules regarding the preparation of research reports and the independence of research analysts. This research report is provided for distribution to "major U.S. institutional investors" and "U.S. institutional investors" in reliance on the exemption from registration provided by Rule 15a-6 of the U.S. Securities Exchange Act of 1934, as amended. When distributing research reports to "U.S. institutional investors," HTI USA will accept the responsibilities for the content of the reports. Any U.S. recipient of this research report wishing to effect any transaction to buy or sell securities or related financial instruments based on the information provided in this research report should do so only through Haitong International Securities (USA) Inc. ("HTI USA"), located at 340 Madison Avenue, 12th Floor, New York, NY 10173, USA; telephone (212) 351 6050. HTI USA is a broker-dealer registered in the U.S. with the U.S. Securities and Exchange Commission (the "SEC") and a member of the Financial Industry Regulatory Authority, Inc. ("FINRA"). HTI USA is not responsible for the preparation of this research report nor for the analysis contained therein. Under no circumstances should any U.S. recipient of this research report contact the analyst directly or effect any transaction to buy or sell securities or related financial instruments directly through HSIPL, HTIRL or HTIJKK. The HSIPL, HTIRL or HTIJKK analyst(s) whose name appears in this research report is not registered or qualified as a research analyst with FINRA and, therefore, may not be subject to FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account. Investing in any non-U.S. securities or related financial instruments (including ADRs) discussed in this research report may present certain risks. The securities of non-U.S. issuers may not be registered with, or be subject to U.S. regulations. Information on such non-U.S. securities or related financial instruments may be limited. Foreign companies may not be subject to audit and reporting standards and regulatory requirements comparable to those in effect within the U.S. The value of any investment or income from any securities or related financial instruments discussed in this research report denominated in a currency other than U.S. dollars is subject to exchange rate fluctuations that may have a positive or adverse effect on the value of or income from such securities or related financial instruments. All inquiries by U.S. recipients should be directed to:

Haitong International Securities (USA) Inc.
340 Madison Avenue, 12th Floor
New York, NY 10173
Attn: Sales Desk at (212) 351 6050

People's Republic of China (PRC): In the PRC, the research report is directed for the sole use of those who receive the research report in accordance with the applicable PRC laws and regulations. Further, the information on the research report does not constitute "production and business activities in the PRC" under relevant PRC laws. This research report does not constitute a public offer of the security, whether by sale or subscription, in the PRC. Further, no legal or natural persons of the PRC may directly or indirectly purchase any of the security or any beneficial interest therein without obtaining all prior PRC government approvals or licenses that are required, whether statutorily or otherwise. Persons who come into possession of this research are required to observe these restrictions.

Notice to Canadian Investors: Under no circumstances is this research report to be construed as an offer to sell securities or as a solicitation of an offer to buy securities in any jurisdiction of Canada. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by Haitong International Securities (USA) Inc., a dealer relying on the "international dealer exemption" under National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations ("NI 31-103") in Alberta, British Columbia, Ontario and Quebec. This research report is not, and under no circumstances should be construed as, a prospectus, an offering memorandum, an advertisement or a public offering of any securities in Canada. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon this research report, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. Upon receipt of this research report, each Canadian recipient will be deemed to have represented that the investor is an "accredited investor" as such term is defined in section 1.1 of National Instrument 45-106 Prospectus Exemptions or, in Ontario, in section 73.3(1) of the Securities Act (Ontario), as applicable, and a "permitted client" as such term is defined in section 1.1 of NI 31-103, respectively.

Notice to Singapore investors: This research report is provided in Singapore by or through Haitong International Securities (Singapore) Pte Ltd ("HTISSPL") [Co Reg No 201306644N. HTISSPL is an Exempt Financial Adviser under the Financial Advisers Act (Cap. 110) ("FAA") to (a) advise on securities, units in a collective investment scheme, exchange-traded derivatives contracts and over-the-counter derivatives contracts and (b) issue or promulgate research analyses or research reports on securities, exchange-traded derivatives contracts and over-the-counter derivatives contracts. This research report is only provided to institutional investors, within the meaning of Section 4A of the Securities and Futures Act (Cap. 289). Recipients of this research report are to contact HTISSPL via the details below in respect of any matters arising from, or in connection with, the research report:

Haitong International Securities (Singapore) Pte. Ltd.
10 Collyer Quay, #19-01 - #19-05 Ocean Financial Centre, Singapore 049315
Telephone: (65) 6536 1920

Notice to Japanese investors: This research report is distributed by Haitong International Securities Company Limited and intended to be distributed to Financial Services Providers or Registered Financial Institutions engaged in investment management (as defined in the Japan Financial Instruments and Exchange Act ("FIEL") Art. 61(1), Order for Enforcement of FIEL Art. 17-11(1), and related articles).

Notice to UK and European Union investors: This research report is distributed by Haitong International Securities Company Limited. This research is directed at persons having professional experience in matters relating to investments. Any investment or investment activity to which this research relates is available only to such persons or will be engaged in only with such persons. Persons who do not have professional experience in matters relating to investments should not rely on this research. Haitong International Securities Company Limited's affiliates may have a net long or short financial interest in excess of 0.5% of the total issued share capital of the entities mentioned in this research report. Please be aware that any report in English may have been published previously in Chinese or another language.

Notice to Australian investors: The research report is distributed in Australia by Haitong International Securities (Singapore) Pte Ltd, Haitong International Securities Company Limited, and Haitong International Securities (UK) Limited in reliance on ASIC Class Order 03/1102, 03/1103 or 03/1099, respectively, which exempts those HTISG entities from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the financial services it provides to wholesale clients in Australia. A copy of the ASIC Class Orders may be obtained at the following website, www.legislation.gov.au. Financial services provided by Haitong International Securities (Singapore) Pte Ltd, Haitong International Securities Company Limited, and Haitong International Securities (UK) Limited are regulated under foreign laws and regulatory requirements, which are different from the laws applying in Australia.

Notice to Indian investors: The research report is distributed by Haitong Securities India Private Limited ("HSIPL"), an Indian company and a Securities and Exchange Board of India ("SEBI") registered Stock Broker, Merchant Banker and Research Analyst that, inter alia, produces and distributes research reports covering listed entities on the BSE Limited ("BSE") and the National Stock Exchange of India Limited ("NSE") (collectively referred to as "Indian Exchanges").

This research report is intended for the recipients only and may not be reproduced or redistributed without the written consent of an authorized signatory of HTISG.

Copyright: Haitong International Securities Group Limited 2019. All rights reserved.

<http://equities.htisec.com/x/legal.html>